

Samuel (Min-Hsuan) Yeh

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Research Interest

Reliable machine learning and Natural language processing

Education

- 2024– *Ph.D. in Computer Science*
University of Wisconsin-Madison, Madison, WI
• Advisor: [Prof. Sharon Li](#)
- 2022–24 *M.S. in Computer Science*
University of Massachusetts-Amherst, Amherst, MA
- 2016–20 *B.Sc. in Computer Science*
National Taiwan University of Science and Technology, Taipei, Taiwan

Research Experience

- 2026 *Research Assistant*
Microsoft Research, Madison, WI, USA (Supervisor: [Shaleen Deep](#))
• Research on *failure attribution for AI agent*
- 2025 *W.J. Cody Associates*
Argonne National Laboratory, Lemont, IL, USA (Supervisor: [Dr. Tanwi Mallick](#))
• Research on *hallucination detection*
- 2023–24 *Research Assistant Intern*
[Crowd-AI Lab](#), PSU, University Park, PA, USA (Advisor: [Prof. Ting-Hao Huang](#))
• Research on *logical fallacy detection*
- 2022–23 *Independent Study*
UMass-Amherst, Amherst, MA, USA (Advisor: [Prof. Philip S. Thomas](#))
• Research on *fair machine learning*
- 2020–22 *Research Assistant*
[NLPSA Lab](#), Academia Sinica, Taipei, Taiwan (Advisor: [Dr. Lun-Wei Ku](#))
• Research on *visual question generation and lie detection*
- 2017–20 *Undergraduate Research Assistant*
[GAME Lab](#), NTUST, Taipei, Taiwan (Advisor: [Prof. Wen-Kai Tai](#))
• Research on *game design and reinforcement learning*

Publications

 [Google Scholar](#)

Peer-reviewed Conference Proceedings

19. **Yeh, Samuel** & Li, S. *How Retrieved Context Shapes Internal Representations in RAG* in *Findings of the Association for Computational Linguistics: ACL 2026* (July 2026).
18. Oh, C., Park, S., Kim, T. E., Li, J., Li, W., **Yeh, Samuel**, Du, S., Hassani, H., Bogdan, P., Song, D. & Li, S. *Uncertainty Quantification in LLM Agents: Foundations, Emerging Challenges, and Opportunities* in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2026).
17. **Yeh, Samuel**, Li, S. & Mallick, T. *LUMINA: Detecting Hallucinations in RAG System with Context-Knowledge Signals* in *The Fourteenth International Conference on Learning Representations* (2026).
16. Xu, Y., Zhang, X., **Yeh, Samuel**, Dhamala, J., Dia, O., Gupta, R. & Li, S. *LH-DECEPTION: Simulating and Understanding LLM Deceptive Behaviors in Long-Horizon Interactions* in *The Fourteenth International Conference on Learning Representations* (2026).
15. **Yeh, Samuel** & Li, S. *Clean First, Align Later: Benchmarking Preference Data Cleaning for Reliable LLM Alignment* in *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track* (2025).
14. Zhang, X., Chen, Y., **Yeh, Samuel** & Li, S. *MetaMind: Modeling Human Social Thoughts with Metacognitive Multi-Agent Systems* in *The Thirty-ninth Annual Conference on Neural Information Processing Systems* (2025).
13. **Yeh, Min-Hsuan**, Wang, J., Du, X., Park, S., Tao, L., Im, S. & Li, Y. *Position: Challenges and Future Directions of Data-Centric AI Alignment* in *Forty-second International Conference on Machine Learning Position Paper Track* (2025).
12. Park, S., **Yeh, Min-Hsuan**, Du, X. & Li, Y. *How to Steer LLM Latents for Hallucination Detection?* in *Forty-second International Conference on Machine Learning* (2025).
11. **Yeh, Min-Hsuan**, Wan, R. & Huang, T.-H. *CoCoLoFa: A Dataset of News Comments with Common Logical Fallacies Written by LLM-Assisted Crowds* in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing* (2024).
10. **Yeh, Min-Hsuan**, Metevier, B., Hoag, A. & Thomas, P. *Analyzing the Relationship Between Difference and Ratio-Based Fairness Metrics* in *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency* (2024).
9. **Yeh, Min-Hsuan**, Chen, V., Huang, T.-H. & Ku, L.-W. *Multi-VQG: Generating Engaging Questions for Multiple Images* in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing* (2022).
8. **Yeh, Min-Hsuan** & Ku, L.-W. *Lying Through One's Teeth: A Study on Verbal Leakage Cues* in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (2021).

Journal Articles

7. **Yeh, Min-Hsuan**, Kamachee, M., Park, S. & Li, Y. *HalluEntity: Benchmarking and Understanding Entity-Level Hallucination Detection*. *Transactions on Machine Learning Research*. ISSN: 2835-8856 (2025).

Working papers

6. **Yeh, Samuel**, Zhu, Y., Deep, S. & Li, S. *Tracing Agentic Failure from the Flow of Success* Under Review. 2026.
5. Cai, L., **Yeh, Samuel**, Dhamala, J., Gupta, R. & Li, S. *DECOR: Auditing LLM Deception via Information Manipulation Theory* Under Review. 2026.
4. Li, J., **Yeh, Samuel** & Li, S. *Multi-Head Recurrent Memory Agents* Under Review. 2026.
3. Oh, C., Wendi Li, S. P., **Yeh, Samuel**, Mallick, T. & Li, S. *Neglected Free Lunch from Post-training: Progress Advantage for LLM Agents* Under Review. 2026.
2. Park, S., Li, W., Oh, C., **Yeh, Samuel**, Kira, Z., Hagenow, M. & Li, S. *Hide-and-Seek in Trajectories: Discovering Failure Signals for VLA Runtime Monitoring* Under Review. 2026.
1. Zhang, X., Chen, Y., **Yeh, Samuel** & Li, S. *Cognition-of-Thought Elicits Social-Aligned Reasoning in Large Language Models* Under Review. 2025.

Academic Service

Reviewer

NeurIPS	'26
ICLR	'25, '26
ACL	'24, '25, '26
COLING	'24, '25
AAAI	'22

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